

LEMMO: Language–Electromagnetic Modeling for Long IQ Recordings

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Abstract

Understanding electromagnetic recordings through language requires retaining evidence over long observation intervals and associating it with individual signals and their relations. High sampling rates make long recordings costly to encode, while detailed descriptions require more information than a single class label. We introduce LEMMO, a signal–language framework for recognition and detailed description from raw in-phase and quadrature (IQ) sequences. A linear-attention encoder pretrained by masked waveform reconstruction processes signals of different lengths. For long recordings, anti-aliased temporal resampling compresses consecutive chunks, followed by bidirectional fusion across chunks. A question-conditioned query bridge maps these features to a fixed set of language-model tokens. The recognition configuration uses compact queries, while the description configuration organizes queries into global, object, and relation groups. Physical-time coordinates provide a temporal reference, and auxiliary fact objectives supervise attributes, object-to-token associations, and temporal or spectral relations. The description configuration is designed to accommodate up to eight million IQ samples and supplies 128 signal tokens to the language model.

1 Introduction

Connecting electromagnetic signals to language models extends multimodal understanding to the waveforms that carry wireless transmissions. A useful interface should answer questions about both signal identity and the structure of a recorded scene. For a recording with multiple transmissions, this includes identifying signal components, describing their physical attributes, and relating their behavior over time and frequency. Such descriptions require signal evidence at several temporal scales. A brief observation can contain millions of samples, yet the language model must receive a compact representation that retains the information relevant to the question.

Existing approaches connect signals to language through either visual representations or features extracted from raw waveforms. Spectrogram-based models reuse visual encoders to answer questions about radio signals [11, 12]. Magnitude spectrograms expose time–frequency structure, but omit phase information from the complex transform. Raw in-phase and quadrature (IQ) sequences instead represent each complex baseband sample by two real-valued components, retaining amplitude and phase before subsequent processing. Signal encoders such as EMind learn reusable representations from IQ data [6], while MERLIN aligns signal representations with an LLM for question answering [9]. These approaches establish a basis for learning from signals, but input length remains a central design consideration.

The length of an IQ sequence reflects both the sampling rate and the physical observation interval. At a sampling rate of 100 MHz, for example, a 40 ms observation contains four million complex samples. A short crop from such a recording may capture a local waveform while missing recurring bursts, frequency changes, or interactions between transmissions. Simply increasing the input length also raises the cost of encoding the signal and passing its features to an LLM. The problem therefore requires both efficient signal encoding and compression that retains temporal organization across the observation.

Detailed description introduces a complementary requirement: the representation must separate the evidence for different signal components and their relations. Recent models already go beyond label

prediction. RF-GPT produces explanations and structured answers, and PReD combines signal perception and recognition with decision-oriented responses [4, 11]. For detailed descriptions of long recordings, however, generating more text is insufficient. The model needs to associate attributes with the correct signal component and express temporal or spectral relations between components. This motivates an interface that both retrieves evidence across a long observation and organizes that evidence for language generation.

We introduce LEMMO, a signal–language architecture for fixed-label recognition and detailed signal description. Pretrained by masked waveform reconstruction, a linear-attention encoder uses a shared patch embedding to process IQ sequences of different lengths. Segment-aware packing improves utilization for short records while keeping unrelated signals separate. For long recordings, temporal resampling compresses consecutive chunks through anti-aliased downsampling, followed by fusion across chunks in chronological order. The description configuration is designed to accommodate up to eight million IQ samples. This capacity refers to the raw input before compression; the language model receives a fixed set of signal embeddings through a question-conditioned query bridge.

The bridge uses 16 queries for recognition and 128 queries for detailed description. The description queries are organized into global, object, and relation groups. Physical-time coordinates provide a temporal reference after compression. Auxiliary fact objectives supervise signal attributes, object-to-token associations, and temporal or spectral relations. All query embeddings are passed to the language model for generation. The two configurations share the signal tokenization and encoder design, with different query capacities for their respective outputs. Together, long-signal resampling and structured queries address two linked requirements: extending the observation available to the model and organizing its contents into descriptions of signal objects, attributes, and relations.

2 Related Work

Existing models for electromagnetic signals fall into two primary categories based on the capability of natural language dialogue: electromagnetic signal encoders and electromagnetic signal large models. This section investigates the development of both categories.

2.1 Electromagnetic Signal Encoders

Electromagnetic signal encoders primarily utilize self-supervised learning to extract features from unlabeled data. Initial studies investigate the processing of raw in-phase and quadrature data to preserve critical physical properties. For example, the RIS-MAE model utilizes masked autoencoders directly on raw sequences to retain amplitude and phase information [5]. To extract spatial and temporal structures without heavy preprocessing, the IQFM model employs contrastive learning and task-aware augmentations, such as cyclic time shifting, on multiple-input multiple-output streams [7].

To process complex dependencies in dynamic environments, versatile architectures incorporate spectrograms and channel state information. The 6G WavesFM framework processes diverse inputs using a shared Vision Transformer backbone, achieving significant parameter sharing across downstream tasks [1]. Similarly, the SpectrumFM model designs a hybrid encoder that combines convolutional neural networks and self-attention mechanisms to capture both localized spectral features and long-range dependencies [10].

Handling variable-length signals and reducing computational complexity remain primary technical challenges for long sequences. Approaches utilizing input patching strategies segment long sequences to decrease attention complexity and remove the computational overhead of spectrogram conversion [3]. Furthermore, the EMind model introduces a length-adaptive multi-signal packing mechanism to process heterogeneous signals with lengths ranging from 128 to 4096 samples [6]. While these encoders enhance feature extraction and pre-training efficiency, the models inherently lack the natural language reasoning capabilities required for complex tasks.

2.2 Electromagnetic Signal Large Models

Recent research integrates electromagnetic signals with large language models to enable natural language dialogue and decision-making. Initial frameworks aim to align signal features with expert knowledge. The RadioLLM framework employs hybrid prompt and token reprogramming, complemented by a frequency-attuned fusion module to capture high-frequency details [2]. To mitigate performance degradation in low signal-to-noise ratio scenarios, the MERLIN model introduces a two-stage training paradigm equipped with knowledge distillation [9].

To facilitate visual question answering, recent studies convert raw streams into visual representations. Zou et al. utilize vision language models to process time-series and spectrogram images, significantly improving classification accuracy while generating textual explanations [12]. Building on this approach, the RF-GPT model processes spectrograms via visual encoders and is trained on a massive synthetic dataset to perform tasks such as overlap detection and user counting [11].

To alleviate the computational overhead of continuous signal data, the DiSC-AMC framework maps numerical features to coarse symbolic tokens and utilizes a nearest-neighbor retrieval mechanism, substantially reducing prompt length while maintaining high in-distribution accuracy [8]. Moreover, the PReD model utilizes multi-perspective representations, including raw waveforms and constellation diagrams, to unify perception, recognition, and decision-making tasks, enabling advanced operations such as the formulation of anti-jamming strategies [4].

Despite these advancements, existing large models for electromagnetic signals primarily process short input sequences and focus on multiple-choice questions or brief responses. The capacity to handle extremely long sequences and to generate comprehensive descriptions for complex signal characteristics remains a critical challenge for current architectures.

3 Method

3.1 Overview

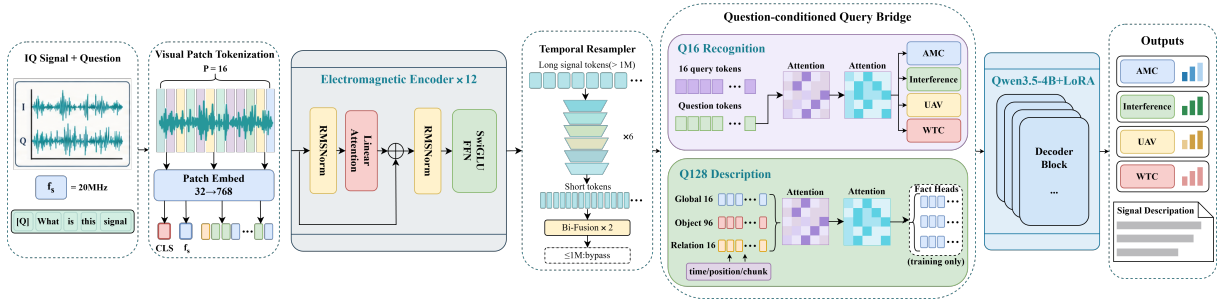


Figure 1: Overview of the proposed framework.

We introduce LEMMO, a unified framework that connects raw complex-baseband electromagnetic signals to both discriminative recognition and natural-language reasoning. The framework is designed to operate across signal records ranging from short communication bursts to multi-million-sample observations.

LEMMO contains three main components: a shared P16 linear-attention signal encoder, a continuity-preserving temporal resampler for long recordings, and a question-conditioned query bridge that converts variable-length signal features into a fixed set of language-model tokens.

Using single-example notation and omitting the batch dimension, given an in-phase/quadrature se-

quence X , sampling rate f_s , and natural-language question q , the framework computes

$$H = R_\psi(E_\theta(X, f_s)), \quad (1)$$

$$H^{(m)} = P_\eta^{(m)}(H), \quad (2)$$

$$T = \text{Emb}_{\text{LM}}(q), \quad (3)$$

$$Z = B_\phi(H^{(m)}, T), \quad (4)$$

where E_θ , R_ψ , and B_ϕ denote the electromagnetic encoder, temporal resampler, and question-conditioned query bridge, respectively. The instance index $m \in \{\text{rec}, \text{desc}\}$ selects compact recognition or structured description.

The compact recognition instance uses 16 queries for modulation, interference, UAV-device, and wireless-technology recognition. The structured description instance uses 128 role-aware queries to represent global scene information, individual signal objects, their attributes, and inter-object relations. Both instances share the same signal representation pathway while using task-aligned query structures for different output granularities.

3.2 Unified Electromagnetic Signal Encoder

Signal normalization and patchification. Let $X_{t,c}$ denote channel $c \in \{1, 2\}$ of sample t . A shared scale factor is applied to the complete I/Q record:

$$s(X) = \max_{t,c} |X_{t,c}|, \quad \bar{X} = \frac{X}{\max\{s(X), \epsilon\}}, \quad (5)$$

where $\epsilon > 0$ ensures numerical stability. Jointly scaling the I and Q channels preserves their relative geometry while reducing acquisition-scale variation.

The normalized sequence is divided into non-overlapping patches of $P = 16$ complex samples. After vectorization, each patch p_i is mapped to a signal embedding:

$$e_i = W_p p_i + b_p. \quad (6)$$

The same patch projection is shared across record lengths and datasets, providing a common signal representation space.

Each signal also receives metadata describing the recording context, including its sampling rate and temporal position. These signals and metadata are jointly encoded so that the resulting IQ-patch features contain both local waveform information and recording-level context.

Linear-attention backbone. To support long IQ sequences, the encoder uses kernelized linear attention with the positive feature map

$$\phi(x) = \text{ELU}(x) + 1. \quad (7)$$

For query, key, and value vectors q_i , k_i , and v_i , the recurrent statistics are

$$S_i = S_{i-1} + \phi(k_i) v_i^\top, \quad (8)$$

$$z_i = z_{i-1} + \phi(k_i), \quad (9)$$

and the output is

$$y_i = \frac{\phi(q_i)^\top S_i}{\phi(q_i)^\top z_i + \epsilon}. \quad (10)$$

This formulation makes computation scale linearly with the number of IQ patches, allowing the same signal encoder to operate on both short and long recordings.

Masked waveform pretraining. The signal encoder is pretrained through masked waveform reconstruction. A subset of IQ patches is hidden, while the visible patches and signal context are encoded. A reconstruction decoder predicts the waveform values at the masked positions.

For each masked patch, the target is standardized independently:

$$\tilde{p}_i = \frac{p_i - \mu(p_i)\mathbf{1}}{\sqrt{\text{Var}(p_i) + \epsilon}}. \quad (11)$$

The reconstruction objective is

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\hat{p}_i - \tilde{p}_i\|_2^2, \quad (12)$$

where $\mathcal{M}$ denotes the masked positions. This objective encourages the encoder to capture local waveform geometry, phase evolution, and contextual predictability. After pretraining, the reconstruction decoder is removed and the encoder is used as the shared signal backbone.

3.3 Continuity-Preserving Long-Signal Resampling

Directly passing all native-resolution signal tokens from a long recording to the language interface is computationally expensive. LEMMO therefore introduces a temporal resampling module that reduces sequence length while retaining the temporal organization of the observation.

Long recordings are processed as chronologically ordered signal regions. Each region is compressed through anti-aliased temporal downsampling followed by adaptive temporal pooling. A generic resampling stage can be written as

$$U^{(\ell+1)} = \text{AA}(U^{(\ell)}) + \tanh(\alpha_\ell) \text{Conv}^{(\ell)}(U^{(\ell)}), \quad (13)$$

where the fixed anti-aliasing branch suppresses aliasing introduced by temporal compression and the learned branch preserves task-relevant transient structure.

The compressed features are then concatenated in chronological order and fused with bidirectional linear attention. For a sequence $U = \{u_j\}_{j=1}^M$, the non-causal kernelized attention is

$$\text{Attn}(q_i, U) = \frac{\phi(q_i)^\top \left(\sum_{j=1}^M \phi(k_j) v_j^\top \right)}{\phi(q_i)^\top \left(\sum_{j=1}^M \phi(k_j) \right) + \epsilon}. \quad (14)$$

The combination of local anti-aliased resampling and global bidirectional fusion preserves temporal continuity during compression while allowing distant parts of the recording to exchange context. The structured-description configuration is designed to accommodate observations containing up to eight million IQ samples.

3.4 Question-Conditioned Signal–Language Interface

Let T contain the language-model embeddings of the question tokens, and let $H^{(m)}$ contain the task-ready signal features. The bridge initializes a set of learnable queries Q_0 and first conditions them on the natural-language question:

$$Q_q = Q_0 + \text{MHA}(\text{LN}(Q_0), \text{LN}(T)W_d^\top, \text{LN}(T)W_d^\top), \quad (15)$$

$$Q_q \leftarrow Q_q + \text{FFN}(\text{LN}(Q_q)). \quad (16)$$

The question-conditioned queries then retrieve signal evidence relevant to the requested output:

$$Q_s = Q_q + \text{MHA}(\text{LN}(Q_q), \text{LN}(H^{(m)}), \text{LN}(H^{(m)})), \quad (17)$$

$$Z = \text{LN}(Q_s + \text{FFN}(\text{LN}(Q_s))) W_o^\top. \quad (18)$$

The resulting query sequence Z is inserted into the language-model input as independent signal tokens. Keeping the queries separate rather than averaging them into a single representation allows the language model to access complementary signal regions, attributes, and objects during generation.

3.5 Compact Recognition Instance

The compact recognition instance uses 16 queries and augments the shared query bridge with task-specific residual adaptation. For recognition task $t \in \mathcal{T}_{\text{rec}}$, the adapted queries are

$$Q^{(t)} = Q_s + \text{Adapter}^{(t)}(Q_s). \quad (19)$$

Here, $\mathcal{T}_{\text{rec}}$ contains modulation, interference, UAV-device, and wireless-technology recognition. The task-specific adaptation allows the shared electromagnetic evidence to be specialized for different recognition objectives without introducing separate signal encoders.

The adapted queries are used in two complementary ways. First, all query tokens are supplied to the language model to support open-ended natural-language answers. Second, an attention-based aggregation of the same queries is passed to a task-specific classifier to provide explicit recognition supervision.

The recognition objective is

$$\mathcal{L}_{\text{rec}} = \mathcal{L}_{\text{LM}} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}, \quad (20)$$

where $\mathcal{L}_{\text{LM}}$ is the language-generation loss over answer tokens and $\mathcal{L}_{\text{cls}}$ is the classification loss for the corresponding recognition task. This dual readout aligns natural-language generation with task-specific electromagnetic discrimination.

3.6 Structured-Description Instance

Detailed electromagnetic description requires the model to associate generated facts with individual signal components and their physical context. LEMMO therefore augments the resampled signal features with physical temporal coordinates before structured query retrieval.

For signal token i , the coordinate representation contains three types of information: its relative location in the complete recording, its physical time on a logarithmic scale, and its position within the long-record structure. We write this coordinate abstractly as

$$\rho_i = [\rho_i^{\text{rel}}, \rho_i^{\text{time}}, \rho_i^{\text{region}}]. \quad (21)$$

The coordinates are encoded with fixed Fourier features and a lightweight projection network, and are added to the corresponding signal feature:

$$\tilde{h}_i = \text{LN}(h_i + \text{PosEnc}(\rho_i)). \quad (22)$$

Because the coordinates are defined with respect to the original observation, they provide a physical temporal reference even after signal compression.

The structured bridge organizes its 128 queries as

$$128 = 16_{\text{global}} + 16_{\text{objects}} \times 6_{\text{roles}} + 16_{\text{relation}}. \quad (23)$$

Global queries represent scene-level properties and signal counts. The ordered object-query groups represent individual signal components through complementary semantic roles, while relation queries model temporal and spectral interactions among components. Query-group, object-index, and object-role embeddings give the queries persistent semantic identities before question and signal cross-attention.

All structured queries are projected into the language space and provided to the language model. Their organization therefore introduces semantic structure without collapsing the underlying signal evidence.

Structured fact supervision. Auxiliary fact objectives supervise the same structured query representation used for language generation. They cover scene-level properties, object existence and counts, signal attributes, object-to-signal associations, and temporal or spectral relations between objects.

For each signal object, its role queries are aggregated into an object representation. Object-to-token association scores are then computed between the object representation and the temporally indexed signal features:

$$s_{o,i} = \text{sim}(o_o, \tilde{h}_i), \quad b_{o,i} = \frac{\exp(s_{o,i})}{\sum_j \exp(s_{o,j})}. \quad (24)$$

This distribution encourages each object query group to retrieve evidence from the corresponding region of the signal record.

For an answer template a , let $\mathcal{A}(a)$ denote the set of structured facts supported by its annotations. The description objective is

$$\mathcal{L}_{\text{desc}} = \mathcal{L}_{\text{LM}} + \sum_{r \in \mathcal{A}(a)} \lambda_r \mathcal{L}_{\text{fact}}^{(r)}. \quad (25)$$

The auxiliary objectives supervise categorical and numerical attributes, object associations, ordering, and temporal or spectral relations when those facts are available in the reference annotation. In this way, structured supervision organizes the signal-to-language representation around objects, attributes, locations, and relations.

4 Experiments

4.1 Datasets and Experimental Setup

Datasets. Table 1 summarizes the four datasets used in our experiments. **LEMMO-ShortQA** is our four-domain short-answer dataset. It covers modulation recognition, interference recognition, UAV-device recognition, and wireless-technology recognition over 45 classes, with raw IQ lengths spanning from 128 to five million complex samples. **LEMMO-LongDesc** is our long-form electromagnetic description dataset. It contains 4G LTE, 5G NR, basic-modulation, ISM, and UAV recordings and pairs long IQ observations with both targeted questions and detailed scene descriptions. Its inputs range from approximately 0.4 to eight million complex samples.

EM-134K [9] provides 134,107 public signal–language pairs for electromagnetic instruction tuning. EM-Bench [9] contains 14 tasks with 300 examples per task: ten perception tasks and four strategy-generation tasks. It is used only for evaluation.

Table 1: Summary of the datasets. Signal counts refer to distinct IQ records or fingerprints. M denotes one million complex samples.

Dataset	Train signal/QA	Dev/Test signal/QA	IQ length	Scope
LEMMO-ShortQA	3,832 / 15,328	870 / 3,480	128–5M	4 recognition tasks
LEMMO-LongDesc	16,103 / 165,328	856 / 8,813	0.4M–8M	5 signal families
EM-134K	95,679 / 120,829	10,587 / 13,278	512/1,024	14 QA tasks
EM-Bench	–	3,370 / 4,200	512/1,024	14 benchmark tasks

Compared models and evaluation. LEMMO couples the proposed IQ representation pathway with Qwen3.5-4B and keeps the pretrained language-model parameters fixed during LoRA adaptation. For short public recordings, all temporally ordered signal features are retained at the language interface. For heterogeneous and long recordings, LEMMO uses the task-aligned interfaces introduced in Sec. 3.4.

We construct **Merlin-LoRA** from the released MERLIN signal encoder, projector, and training pipeline, while replacing full-parameter language-model tuning with LoRA adaptation of a frozen Qwen3-4B-Instruct-2507 backbone. This matches the parameter-efficient language-adaptation policy used by LEMMO while preserving the signal pathway of the public baseline. Both models are trained on the same fingerprint-disjoint EM-134K partition and evaluated on the same EM-Bench examples. Perception tasks use option accuracy and generation tasks use ROUGE-L F1. General-language retention is measured on all 14,042 test questions from the 57-subject, five-shot MMLU benchmark.

4.2 EM-Bench Results and General-Knowledge Retention

The released MERLIN checkpoint adapts all Qwen3 parameters. This maximizes its capacity for electromagnetic specialization, but changes the behavior of the underlying language model substantially: on

our uniform MMLU evaluation, its accuracy decreases from 57.41% for the original Qwen3 backbone to 30.00%. Merlin-LoRA provides the corresponding parameter-efficient comparison, in which the language backbone remains frozen just as in LEMMO.

Table 2: EM-Bench results (%). Perception tasks report accuracy; generation tasks report ROUGE-L F1. The better result in each row is bold.

Task	Metric	LEMMO	Merlin-LoRA
<i>Perception</i>			
CJR	Acc.	83.67	72.33
MOD	Acc.	36.00	27.00
Radar bandwidth	Acc.	58.33	27.00
Radar duty cycle	Acc.	54.33	30.67
Radar number of pulses	Acc.	82.00	81.00
Radar PRF	Acc.	41.33	26.33
Radar pulse width	Acc.	37.33	30.33
PI	Acc.	79.67	71.00
RJR	Acc.	73.67	45.67
SD	Acc.	46.67	24.00
Perception average	Acc.	59.30	43.53
<i>Generation</i>			
Anti-CJ	R-L	43.84	48.76
Anti-RJ	R-L	38.14	33.19
CJS	R-L	37.64	42.92
RJS	R-L	54.71	40.97
Generation average	R-L	43.58	41.46

Table 2 reports all 14 EM-Bench tasks. LEMMO obtains the best result on every perception task. Its ten-task perception average is 59.30%, compared with 43.53% for Merlin-LoRA, an improvement of 15.77 percentage points. LEMMO also raises the four-task generation average from 41.46 to 43.58 ROUGE-L, and leads on 12 of the 14 task-level primary metrics. The gains cover modulation, radar-parameter estimation, protocol identification, jamming recognition, signal detection, and strategy generation, demonstrating broad transfer rather than specialization to a single signal family.

Table 3: Five-shot MMLU accuracy (%) over 57 subjects.

Model	STEM	Humanities	Social sci.	Other	Overall
LEMMO	60.04	59.49	72.64	70.71	64.98
Merlin-LoRA	53.60	48.33	70.26	62.92	57.55
MERLIN	28.45	27.97	32.95	31.70	30.00

Electromagnetic specialization does not come at the expense of general question answering. As shown in Table 3, LEMMO reaches 64.98% on MMLU, 7.43 points above Merlin-LoRA and 34.98 points above the released MERLIN checkpoint. LEMMO also exceeds its own Qwen3.5-4B initialization by 0.63 points, while Merlin-LoRA remains within 0.14 points of its Qwen3 initialization. Thus, LoRA-based signal alignment preserves the general knowledge of both backbones, while LEMMO attains the strongest absolute result in every MMLU domain.

4.3 Short-Answer Electromagnetic Understanding

We next evaluate recognition from naturally phrased, open-ended questions on LEMMO-ShortQA. Table 4 shows that LEMMO consistently outperforms Merlin-LoRA across all four domains. LEMMO achieves 79.93% on average, improving the baseline by 18.31 percentage points. The largest gain occurs for UAV-device recognition, where five-million-sample observations benefit directly from retaining evidence

across the recording. Strong gains on wireless-technology and interference recognition further show that the same model can resolve protocol-level and waveform-level distinctions through a unified natural-language interface.

Table 4: Open-ended recognition accuracy on LEMMO-ShortQA (%).

Task	LEMMO	Merlin-LoRA	Δ
Modulation recognition	82.41	74.94	+7.47
Interference recognition	75.74	67.40	+8.33
UAV-device recognition	85.00	45.62	+39.38
Wireless-technology recognition	76.56	58.52	+18.05
Average	79.93	61.62	+18.31

4.4 Ablation Studies

Table 5: Signal-input intervention on public electromagnetic perception (%).

Input condition	Perception accuracy
Correct IQ	50.93
Task-matched shuffled IQ	28.10
Zero IQ	25.40
Text only	1.97

Dependence on signal evidence. To verify that electromagnetic predictions are grounded in the input waveform, we intervene on the signal supplied to a fixed LEMMO configuration while leaving every question unchanged. Table 5 compares the correct IQ record with a task-matched shuffled record, an all-zero record, and a text-only input. Correct IQ raises perception accuracy by 22.83 points over the shuffled control and by 25.53 points over the zero-signal control. The +48.96-point margin over text-only inference further confirms that task prompts alone cannot account for LEMMO’s electromagnetic decisions.

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